# Walkaround evidence sheet

Two walkaround videos of one rental car, compared panel by panel. Code decides every verdict; a model answer can only lower one.

**evidence support, not legal advice**

Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS. License: https://creativecommons.org/licenses/by/4.0/

| | |
|---|---|
| **run id** | `pairC_core3` |
| **app commit** | `f1fb68bc25ca8a1106a8eded6f5f49b1d3096bdb` |
| **config hash** | `89dbc3e3f49f0557e848db8e744a679f5a8615224e031eacf26d813416e9407f` |
| **events head hash** | `98fe5dddb3261d4f9deedbb80026210b102caeb327bb444faf69592e4ad8f3f8` |
| **rental agreement number** | _____ (fill in) |
| **plate** | plate blurred |

### The three clocks

| sealed at | first model call at | sealed after the camera time |
|---|---|---|
| **2026-11-01T05:19:38.673Z** | **2026-11-01T05:20:10Z** | **camera time not present** |

### Originals

| walk | file | bytes | sha256 | camera says | received | hash on your phone = hash on receipt | proven to exist by |
|---|---|---|---|---|---|---|---|
| pickup | uco3d_7336_narrated.mp4 | `96182720` | `25f68a8992c980976df1352c0d2894eca301a13b576bcac397e61952e1440efb` | not present | 2026-11-01T05:19:38.420Z | match | 2026-11-01T05:19:42Z (RFC 3161 token from freetsa); OpenTimestamps pending at the calendars (ots upgrade adds the Bitcoin block after a few hours) |
| return | uco3d_50152-5241-46685.mp4 | `107096877` | `f49b432f2932d4a9ffeb59881c992c5054e09a2d7d6b816a41d5dd1df9c8646b` | not present | 2026-11-01T05:19:38.533Z | match | 2026-11-01T05:19:42Z (RFC 3161 token from freetsa); OpenTimestamps pending at the calendars (ots upgrade adds the Bitcoin block after a few hours) |

### Privacy

Processed with Meta's Model API in Standard mode, which Meta's documentation says is not used to train Meta models. Uploads were set to expire after one hour and were deleted by Walkaround when the run ended. Model requests were sent with storage off. Meta's terms allow it to retain content for abuse review and legal reasons; Walkaround does not control that. SAM 3.1 saw the unblurred proxy for the length of one segmentation pass to find the plates and faces it then blurred. For the second look, two blurred crops per look were sent to Muse Spark Open on Baseten, a company separate from Meta; no video went there.

Files API upload deleted: 2026-11-01T05:24:23.620Z. Stored responses deleted: none stored or not yet. Quarantine (unblurred originals) deleted: 2026-11-01T05:24:23.620Z.

### What to do with this sheet

1. Start with the dated pickup frames in this sheet: the camera says not present, the file was received 2026-11-01T05:19:38.420Z, and the file hashes (25f68a8992c9, f49b432f2932) are on page 1 and in the sealed manifest.
2. Ask the company in writing for the full damage report, the repair estimate and every image it holds for this rental, including the pickup inspection images; Hertz says customers can request more images.
3. Get an alternative written estimate from a body shop within 10 business days; card network rules compare two quotes (Visa Core Rules Table 5-19; Amex allows up to 115% of the estimate).
4. If the company does not answer, escalate to a named executive and file a complaint with your state consumer protection office; the one sourced renter win came by executive escalation with the company's own scans.
5. Last, the card dispute, with its clocks: 60 days from the statement under Regulation Z 1026.13, 120 days under Visa's rules, and the $50 cap of 15 USC 1666i; a dispute does not stop collections, so keep the paper trail.

### Coverage per walk

**pickup**

- left front door: glimpsed, 1.0 s
- left rear door: not_filmed
- right front door: filmed, 9.2 s
- right rear door: glimpsed, 9.4 s
- left front fender: not_seen
- left rear fender: not_filmed
- right front fender: glimpsed, 3.3 s
- right rear fender: glimpsed, 6.5 s
- hood: filmed, 8.7 s
- roof: glimpsed, 1.2 s
- trunk lid: filmed, 9.4 s
- front bumper: filmed, 7.5 s
- rear bumper: filmed, 6.5 s
- left front wheel: glimpsed, 0.7 s
- left rear wheel: not_filmed
- right front wheel: filmed, 4.5 s
- right rear wheel: filmed, 6.7 s
- headlights: glimpsed, 7.4 s
- tail lights: glimpsed, 7.7 s
- mirrors: glimpsed, 5.6 s
- windshield: glimpsed, 3.9 s
- rear window: filmed, 7.2 s
- missing at pickup: left front wheel, right front wheel, left front fender, left rear fender, right front fender, right rear fender, roof, headlights, tail lights, mirrors, rear window
- top-up asked: Film these again, in this order: left front wheel (glimpsed, 2.6 s); right front wheel (not filmed); left front fender (not seen); left rear fender (glimpsed, 3.9 s); right front fender (not filmed); right rear fender (glimpsed, 2.3 s); roof (glimpsed, 7.5 s); headlights (glimpsed, 9.4 s); tail lights (glimpsed, 14.0 s); mirrors (glimpsed, 7.2 s); rear window (glimpsed, 4.9 s). The app then re-runs 10 of 16 nouns: wheel, fender, roof, headlight, tail light, side mirror, rear window, car, license plate, face. Also film the odometer once.

**return**

- left front door: filmed, 1.9 s
- left rear door: filmed, 3.7 s
- right front door: filmed, 10.7 s
- right rear door: filmed, 10.0 s
- left front fender: not_seen
- left rear fender: glimpsed, 3.9 s
- right front fender: not_filmed
- right rear fender: glimpsed, 2.3 s
- hood: filmed, 12.1 s
- roof: glimpsed, 7.5 s
- trunk lid: filmed, 7.0 s
- front bumper: filmed, 9.7 s
- rear bumper: filmed, 6.1 s
- left front wheel: glimpsed, 2.6 s
- left rear wheel: filmed, 6.9 s
- right front wheel: not_filmed
- right rear wheel: filmed, 8.8 s
- headlights: glimpsed, 9.4 s
- tail lights: glimpsed, 14.0 s
- mirrors: glimpsed, 7.2 s
- windshield: filmed, 10.7 s
- rear window: glimpsed, 4.9 s

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

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## left front door

coverage: pickup 1.0 s (glimpsed); return 1.9 s (filmed)




m3 unsure S
m2 dirt M



pickup: video time 0.27 s



return: video time 19.73 s; marks return_left_front_door_kf1_m1, return_left_front_door_kf1_m2, return_left_front_door_kf1_m3

how it was checked: not compared: keyframes pickup_left_front_door_kf1 with return_left_front_door_kf1 (G1 not filmed, G4 mask IoU: not measured, G6 side uncertain); G1 and G6 are panel gates, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `return_left_front_door_kf1_m1`, `return_left_front_door_kf1_m2`, `return_left_front_door_kf1_m3` `reflection M` | **not comparable** | L6, G1, G4, G6 | no pair | Stays Put: drift 0.54, b1 0.15, b2 0.57, panned; neighbour test: 3 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7
G6 side uncertain: side_uncertain, or front_sign null, on either walk

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## left rear door

coverage: pickup 0.0 s (not_filmed); return 3.7 s (filmed)

no pickup frame for this panel





return: video time 0.00 s; marks return_left_rear_door_kf1_m1, return_left_rear_door_kf1_m2, return_left_rear_door_kf1_m3

## Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| return_left_rear_door_kf1_m1, return_left_rear_door_kf1_m2, return_left_rear_door_kf1_m3<br>scratch M | not comparable | L6, G1 | no pair | Stays Put: no track (SAM did not follow the mark through the clip) |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## right front door

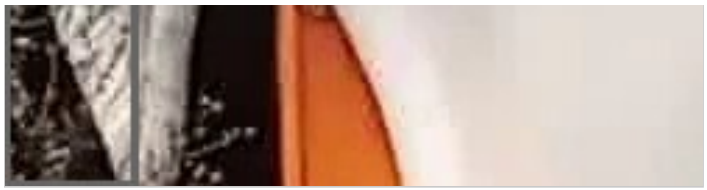
coverage: pickup 9.2 s (filmed); return 10.7 s (filmed)







return: video time 7.13 s; marks return_right_front_door_kf1_m1, return_right_front_door_kf2_m1, return_right_front_door_kf3_m1



pickup: video time 19.20 s; marks pickup_right_front_door_kf1_m1, pickup_right_front_door_kf1_m2

how it was checked: not compared: none of the 3 keyframe pairs passed every gate; tried: pickup_right_front_door_kf1 with return_right_front_door_kf3 (G4 mask IoU 0.64, G9 mask definition differs); pickup_right_front_door_kf1 with return_right_front_door_kf1 (G4 mask IoU 0.66, G9 mask definition differs); pickup_right_front_door_kf1 with return_right_front_door_kf2 (G4 mask IoU 0.64, G9 mask definition differs)

## Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_right_front_door_kf1_m1, pickup_right_front_door_kf1_m2, return_right_front_door_kf1_m1, return_right_front_door_kf2_m1, return_right_front_door_kf3_m1` `unsure M` | **not comparable** | L6, G4, G9 | no pair | Stays Put: drift 0.41, b1 0.08, b2 0.08, panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

L6 not comparable: a G rule failed for the panel; both best frames still show
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7
G9 mask definition differs: window_subtracted differs between the paired keyframes

## What was said

- `5340 to 8060 ms`: front door like scratch near the handle

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## right rear door

coverage: pickup 9.4 s (glimpsed); return 10.0 s (filmed)





return: video time 6.93 s



pickup: video time 19.20 s

how it was checked: not compared: keyframes pickup_right_rear_door_kf1 with return_right_rear_door_kf1 (G1 not filmed, G4 mask IoU 0.58, G9 mask definition differs); G1 is a panel gate, so no other pair was tried

## Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| | not comparable | L6, G1, G4, G9 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7
G9 mask definition differs: window_subtracted differs between the paired keyframes

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## left rear fender

coverage: pickup 0.0 s (not_filmed); return 3.9 s (glimpsed)

no pickup frame for this panel




return: video time 6.47 s; marks return_left_rear_fender_kf3_m1

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `return_left_rear_fender_kf3_m1`<br>`dirt M` | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## right front fender

coverage: pickup 3.3 s (glimpsed); return 0.0 s (not_filmed)



pickup: video time 8.53 s

no return frame for this panel

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## right rear fender

coverage: pickup 6.5 s (glimpsed); return 2.3 s (glimpsed)



pickup: video time 18.40 s; marks pickup_right_rear_fender_kf2_m1, pickup_right_rear_fender_kf3_m1



return: video time 7.73 s

how it was checked: not compared: keyframes pickup_right_rear_fender_kf1 with return_right_rear_fender_kf2 (G1 not filmed, G3 blurry keyframe, G4 mask IoU: panel axes disagree); G1 is a panel gate, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_right_rear_fender_kf2_m1`, `pickup_right_rear_fender_kf3_m1` `dirt S` | not comparable | L6, G1, G3, G4 | no pair | |

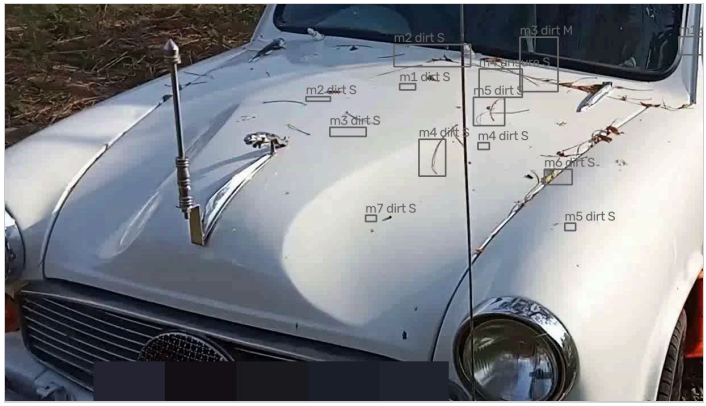
L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

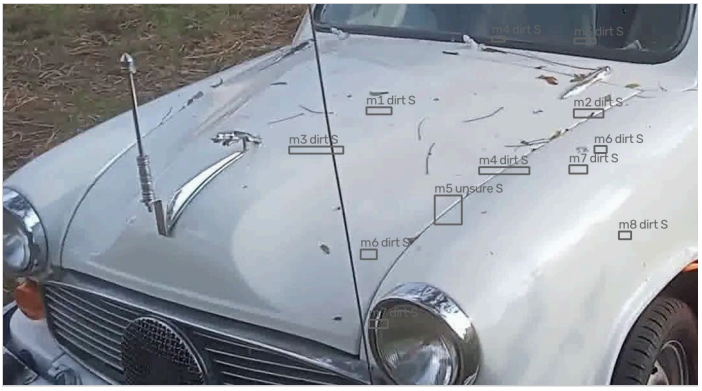
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# hood

coverage: pickup 8.7 s (filmed); return 12.1 s (filmed)




pickup: video time 0.00 s; marks pickup_hood_kf1_m1, pickup_hood_kf1_m2, pickup_hood_kf1_m3, pickup_hood_kf1_m4, pickup_hood_kf1_m5, pickup_hood_kf1_m6, pickup_hood_kf1_m7, pickup_hood_kf2_m3, pickup_hood_kf2_m4, pickup_hood_kf2_m5, pickup_hood_kf2_m6, pickup_hood_kf3_m1, pickup_hood_kf3_m2, pickup_hood_kf3_m3, pickup_hood_kf3_m4, pickup_hood_kf3_m5, pickup_hood_kf3_m6




return: video time 19.20 s; marks return_hood_kf1_m1, return_hood_kf1_m2, return_hood_kf1_m3, return_hood_kf1_m4, return_hood_kf1_m5, return_hood_kf1_m6, return_hood_kf1_m7, return_hood_kf1_m8, return_hood_kf2_m1, return_hood_kf2_m2, return_hood_kf2_m3, return_hood_kf2_m4, return_hood_kf2_m5, return_hood_kf2_m6, return_hood_kf2_m7

how it was checked: not compared: none of the 6 keyframe pairs passed every gate; tried: pickup_hood_kf1 with return_hood_kf2 (G4 mask IoU: panel axes disagree, G8 hood IoU: panel axes disagree); pickup_hood_kf1 with return_hood_kf1 (G4 mask IoU: panel axes disagree, G8 hood IoU: panel axes disagree); pickup_hood_kf2 with return_hood_kf1 (G4 mask IoU 0.70, G8 hood IoU 0.70); pickup_hood_kf2 with return_hood_kf2 (G4 mask IoU 0.66, G8 hood IoU 0.66); pickup_hood_kf3 with return_hood_kf1 (G4 mask IoU: panel axes disagree, G8 hood IoU: panel axes disagree); pickup_hood_kf3 with return_hood_kf2 (G4 mask IoU: panel axes disagree, G8 hood IoU: panel axes disagree)

## Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_hood_kf1_m1, pickup_hood_kf1_m2, pickup_hood_kf1_m3, pickup_hood_kf1_m4, pickup_hood_kf1_m5, pickup_hood_kf1_m6, pickup_hood_kf1_m7, pickup_hood_kf2_m3, pickup_hood_kf2_m4, pickup_hood_kf2_m5, pickup_hood_kf2_m6, pickup_hood_kf3_m1, pickup_hood_kf3_m2, pickup_hood_kf3_m3, pickup_hood_kf3_m4, pickup_hood_kf3_m5, pickup_hood_kf3_m6, return_hood_kf1_m1, return_hood_kf1_m2, return_hood_kf1_m3, return_hood_kf1_m4, return_hood_kf1_m5, return_hood_kf1_m6, return_hood_kf1_m7, return_hood_kf1_m8, return_hood_kf2_m1, return_hood_kf2_m2, return_hood_kf2_m3, return_hood_kf2_m4, return_hood_kf2_m5, return_hood_kf2_m6, return_hood_kf2_m7` `dirt S` | **not comparable** | L6, G4, G8 | no pair | Stays Put: drift 0.38, b1 0.10, b2 0.09, panned; neighbour test: 11.0x its 5 neighbours' wobble over the clip and 1.3x within 0.5 s; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

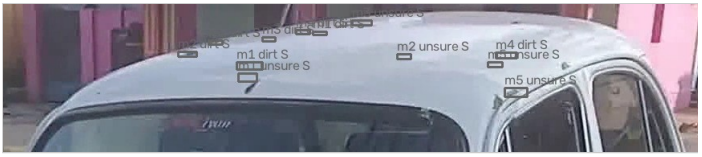
L6 not comparable: a G rule failed for the panel; both best frames still show
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7
G8 hood IoU: hood pair with whitened mask IoU under 0.8

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# roof

coverage: pickup 1.2 s (glimpsed); return 7.5 s (glimpsed)

no pickup frame for this panel




return: video time 19.33 s; marks return_roof_kf1_m1, return_roof_kf1_m2, return_roof_kf1_m3, return_roof_kf1_m4, return_roof_kf1_m5, return_roof_kf2_m1, return_roof_kf2_m2, return_roof_kf2_m3, return_roof_kf2_m4, return_roof_kf2_m5, return_roof_kf3_m1, return_roof_kf3_m2

## Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| return_roof_kf1_m1, return_roof_kf1_m2, return_roof_kf1_m3, return_roof_kf1_m4, return_roof_kf1_m5, return_roof_kf2_m1, return_roof_kf2_m2, return_roof_kf2_m3, return_roof_kf2_m4, return_roof_kf2_m5, return_roof_kf3_m1, return_roof_kf3_m2 dirt S | not comparable | L6, G1, G6 | no pair | Stays Put: drift 0.07, b1 0.03, b2 0.01, not panned; neighbour test: 1 neighbour tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G6 side uncertain: side_uncertain, or front_sign null, on either walk

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## trunk lid

coverage: pickup 9.4 s (filmed); return 7.0 s (filmed)




pickup: video time 14.93 s; marks pickup_trunk_lid_kf1_m1, pickup_trunk_lid_kf1_m2, pickup_trunk_lid_kf1_m3, pickup_trunk_lid_kf2_m2




return: video time 3.27 s; marks return_trunk_lid_kf1_m1, return_trunk_lid_kf1_m2, return_trunk_lid_kf1_m3, return_trunk_lid_kf2_m3

how it was checked: compared on keyframes pickup_trunk_lid_kf2 with return_trunk_lid_kf1, the first pair tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| pickup_trunk_lid_kf1_m1, return_trunk_lid_kf1_m2 dirt S | already there | L3 | 0.09 units, about 3 cm | Stays Put: no track (SAM did not follow the mark through the clip)<br>size not stated: the license plate is turned more than 30 degrees from the camera<br>pickup view flagged dirty: the mark is in the same place in both walks, so it was there at pickup, whatever it is<br>Muse Spark typed both marks dirt (K4), with low confidence on the return mark (K12): a note here, since the mark sits in the same place in both walks and the second look compared the two crops and said same |
| pickup_trunk_lid_kf1_m2 scratch S | needs your eye | L4, L1, K15 | no pair | Stays Put: no track (SAM did not follow the mark through the clip)<br>size not stated: the license plate is turned more than 30 degrees from the camera<br>pickup view flagged dirty: could be dirt |
| pickup_trunk_lid_kf1_m3 dirt S | needs your eye | L4, L1, K15, K4 | no pair | Stays Put: drift 0.05, b1 0.18, b2 0.07, not panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label<br>size not stated: the license plate is turned more than 30 degrees from the camera<br>pickup view flagged dirty: could be dirt |
| pickup_trunk_lid_kf2_m2 unsure S | needs your eye | L4, L1, K12, K4 | no pair | size not stated: the license plate is turned more than 30 degrees from the camera |

| | | | | |
|---|---|---|---|---|
| `return_trunk_lid_kf1_m1`<br>`dirt S` | needs your eye | L4, L2, K12, K4 | no pair | Stays Put: no track (SAM did not follow the mark through the clip)<br>size not stated: the license plate is turned more than 30 degrees from the camera |
| `return_trunk_lid_kf1_m3`<br>`dirt S` | needs your eye | L4, L2, K12, K4 | no pair | size not stated: the license plate is turned more than 30 degrees from the camera |
| `return_trunk_lid_kf2_m3`<br>`unsure S` | needs your eye | L4, L2, K12, K15, K4 | no pair | size not stated: the license plate is turned more than 30 degrees from the camera<br>return view flagged glare: could be a reflection |

L3 already there: d < tau_pre_r
L4 needs your eye: tau_pre_r <= d < tau_eye_r, or any K flag lowered the label
L1 not seen at return: d is None for a pickup mark: not_seen_at_return
K15 view flagged: Muse Spark flagged the pickup or return view of the panel wet, dirty or glare, so a mark seen in one walk only may be dirt, water or a reflection, or may have been hidden in the other walk; a mark in the same place in both walks keeps its label with a note
K4 uncertain type: type dirt, reflection or unsure
K12 low confidence: Spark confidence low on either mark
L2 new or unseen: d is None for a return mark: new_or_unseen

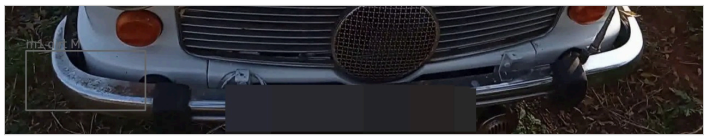
### What was said

- `12940 to 16220 ms`: trunk lid two small chips by the edge

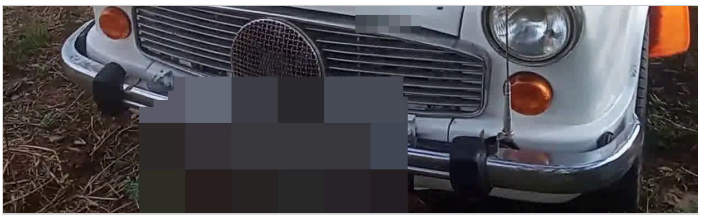
Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

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## front bumper



pickup: video time 3.60 s; marks pickup_front_bumper_kf1_m1



return: video time 17.60 s

coverage: pickup 7.5 s (filmed); return 9.7 s (filmed)

how it was checked: not compared: none of the 2 keyframe pairs passed every gate; tried: pickup_front_bumper_kf1 with return_front_bumper_kf2 (G4 mask IoU 0.48); pickup_front_bumper_kf1 with return_front_bumper_kf1 (G4 mask IoU 0.55)

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_front_bumper_kf1_m1`<br>`dirt M` | **not comparable** | L6, G4 | no pair | Stays Put: drift 0.68, b1 0.18, b2 0.79, not panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

L6 not comparable: a G rule failed for the panel; both best frames still show
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7

### What was said

- `140 to 4300 ms`: pick up work starting at the front front bumper it looks clean

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## rear bumper




pickup: video time 15.53 s; marks pickup_rear_bumper_kf1_m1, pickup_rear_bumper_kf3_m1, pickup_rear_bumper_kf3_m2




return: video time 2.00 s; marks return_rear_bumper_kf1_m1, return_rear_bumper_kf2_m1

coverage: pickup 6.5 s (filmed); return 6.1 s (filmed)

how it was checked: compared on keyframes pickup_rear_bumper_kf1 with return_rear_bumper_kf1; tried first: pickup_rear_bumper_kf1 with return_rear_bumper_kf2 (G4 mask IoU 0.70)

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_rear_bumper_kf1_m1`<br>`dirt S` | needs your eye | L4, L1, K15, K2, K4 | no pair | Stays Put: no track (SAM did not follow the mark through the clip)<br>size not stated: the license plate is turned more than 30 degrees from the camera; no wheel measured on that frame<br>return view flagged dirty and glare: could have hidden the mark |
| `pickup_rear_bumper_kf3_m1`<br>`dirt S` | needs your eye | L4, L1, K12, K15, K2, K4 | no pair | Stays Put: no track (SAM did not follow the mark through the clip)<br>size not stated: the license plate is turned more than 30 degrees from the camera; no wheel measured on that frame<br>pickup view flagged dirty and glare: could be dirt or a reflection; return view flagged dirty and glare: could have hidden the mark |
| `pickup_rear_bumper_kf3_m2`<br>`scuff M` | needs your eye | L4, L1, K12, K15, K2 | no pair | size not stated: the license plate is turned more than 30 degrees from the camera; no wheel measured on that frame<br>Hertz's page lists scuffs that do not break paint as ordinary wear (REV 7.2025, illustrative only; page saved 2026-10-26).<br>pickup view flagged dirty and glare: could be dirt or a reflection; return view flagged dirty and glare: could have hidden the mark |
| `return_rear_bumper_kf1_m1`<br>`dirt S` | needs your eye | L4, L2, K15, K4 | no pair | Stays Put: drift 0.70, b1 0.29, b2 0.51, not panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label<br>size not stated: the license plate is turned more than 30 degrees from the camera; no wheel measured on that frame<br>return view flagged dirty and glare: could be dirt or a reflection |

| | | | | |
|---|---|---|---|---|
| return_rear_bumper_kf2_m1<br>dirt S | needs your eye | L4, L2, K15, K4 | no pair | Stays Put: drift 0.59, b1 0.28, b2 1.15, not panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label<br>size not stated: the license plate is turned more than 30 degrees from the camera; no wheel measured on that frame<br>return view flagged dirty and glare: could be dirt or a reflection |

L4 needs your eye: tau_pre_r <= d < tau_eye_r, or any K flag lowered the label
L1 not seen at return: d is None for a pickup mark: not_seen_at_return
K15 view flagged: Muse Spark flagged the pickup or return view of the panel wet, dirty or glare, so a mark seen in one walk only may be dirt, water or a reflection, or may have been hidden in the other walk; a mark in the same place in both walks keeps its label with a note
K2 single frame: the mark appears in 1 of 3 keyframes while the other two were complete and sharp
K4 uncertain type: type dirt, reflection or unsure
K12 low confidence: Spark confidence low on either mark
L2 new or unseen: d is None for a return mark: new_or_unseen

### What was said

- `16700 to 18380 ms`: rear bumper looks fine

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## left front wheel







return: video time 18.27 s; marks return_left_front_wheel_kf1_m1



pickup: video time 0.33 s; marks pickup_left_front_wheel_kf1_m1

coverage: pickup 0.7 s (glimpsed); return 2.6 s (glimpsed)

how it was checked: not compared: keyframes pickup_left_front_wheel_kf1 with return_left_front_wheel_kf1 (G1 not filmed); G1 is a panel gate, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_left_front_wheel_kf1_m1, return_left_front_wheel_kf1_m1` `unsure S` | **not comparable** | L6, G1 | no pair | Stays Put: no track (SAM did not follow the mark through the clip) |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## left rear wheel

no pickup frame for this panel



return: video time 6.87 s

coverage: pickup 0.0 s (not_filmed); return 6.9 s (filmed)

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

## right front wheel



no return frame for this panel

pickup: video time 7.87 s

coverage: pickup 4.5 s (filmed); return 0.0 s (not_filmed)

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## right rear wheel



pickup: video time 10.80 s



return: video time 10.20 s

coverage: pickup 6.7 s (filmed); return 8.8 s (filmed)

how it was checked: compared on keyframes pickup_right_rear_wheel_kf1 with return_right_rear_wheel_kf2, the first pair tried

No marks compared on this panel.

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## headlights

coverage: pickup 7.4 s (glimpsed); return 9.4 s (glimpsed)



pickup: video time 3.33 s




return: video time 17.87 s; marks return_headlights_kf1_m1, return_headlights_kf1_m2, return_headlights_kf2_m1

how it was checked: not compared: keyframes pickup_headlights_kf2 with return_headlights_kf1 (G1 not filmed); G1 is a panel gate, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `return_headlights_kf1_m1, return_headlights_kf1_m2, return_headlights_kf2_m1` `reflection L` | not comparable | L6, G1 | no pair | Stays Put: drift 1.11, b1 0.38, b2 0.11, not panned; neighbour test: 0 neighbours tracked through the clip, 4 needed; inconclusive; recorded only: Stays Put is not calibrated yet and changes no label |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## tail lights

coverage: pickup 7.7 s (glimpsed); return 14.0 s (glimpsed)



pickup: video time 19.20 s



return: video time 7.80 s

how it was checked: not compared: keyframes pickup_tail_lights_kf1 with return_tail_lights_kf2 (G1 not filmed); G1 is a panel gate, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## mirrors

coverage: pickup 5.6 s (glimpsed); return 7.2 s (glimpsed)



pickup: video time 6.67 s; marks pickup_mirrors_kf1_m1



return: video time 8.67 s; marks return_mirrors_kf2_m1

how it was checked: not compared: keyframes pickup_mirrors_kf2 with return_mirrors_kf3 (G1 not filmed, G3 blurry keyframe, G4 mask IoU: not measured, G6 side uncertain); G1 and G6 are panel gates, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_mirrors_kf1_m1, return_mirrors_kf2_m1` `reflection M` | **not comparable** | L6, G1, G3, G4, G6 | no pair | Stays Put: no track (SAM did not follow the mark through the clip) |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe
G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7
G6 side uncertain: side_uncertain, or front_sign null, on either walk

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Piper TTS. f49b432f2932

## windshield

coverage: pickup 3.9 s (glimpsed); return 10.7 s (filmed)

no pickup frame for this panel



return: video time 18.07 s; marks return_windshield_kf1_m1, return_windshield_kf1_m2

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| return_windshield_kf1_m1, return_windshield_kf1_m2<br>reflection L | not comparable | L6, G1 | no pair | |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk

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## rear window

coverage: pickup 7.2 s (filmed); return 4.9 s (glimpsed)




pickup: video time 18.73 s; marks pickup_rear_window_kf1_m1, pickup_rear_window_kf1_m2, pickup_rear_window_kf1_m3, pickup_rear_window_kf1_m4, pickup_rear_window_kf2_m2, pickup_rear_window_kf2_m4




return: video time 0.07 s; marks return_rear_window_kf1_m1, return_rear_window_kf1_m2, return_rear_window_kf1_m3, return_rear_window_kf1_m4, return_rear_window_kf2_m1, return_rear_window_kf2_m3, return_rear_window_kf2_m4

how it was checked: not compared: keyframes pickup_rear_window_kf1 with return_rear_window_kf2 (G1 not filmed, G3 blurry keyframe); G1 is a panel gate, so no other pair was tried

### Verdicts

| marks | verdict | rule ids | distance | detail |
|---|---|---|---|---|
| `pickup_rear_window_kf1_m1, pickup_rear_window_kf1_m2, pickup_rear_window_kf1_m3, pickup_rear_window_kf1_m4, pickup_rear_window_kf2_m2, pickup_rear_window_kf2_m4, return_rear_window_kf1_m1, return_rear_window_kf1_m2, return_rear_window_kf1_m3, return_rear_window_kf1_m4, return_rear_window_kf2_m1, return_rear_window_kf2_m3, return_rear_window_kf2_m4`<br>`crack L` | **not comparable** | L6, G1, G3 | no pair | Stays Put: no track (SAM did not follow the mark through the clip) |

L6 not comparable: a G rule failed for the panel; both best frames still show
G1 not filmed: the panel is not 'filmed' in either walk
G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe

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## Panels without frames

| panel | coverage |
|---|---|
| left front fender | pickup: not_seen (0.0 s); return: not_seen (0.0 s) |

## Not comparable

| panel | rule ids | why |
|---|---|---|
| left front door | G1, G4, G6 | G1 not filmed: the panel is not 'filmed' in either walk<br>G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7<br>G6 side uncertain: side_uncertain, or front_sign null, on either walk |
| left rear door | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| right front door | G4, G9 | G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7<br>G9 mask definition differs: window_subtracted differs between the paired keyframes |
| right rear door | G1, G4, G9 | G1 not filmed: the panel is not 'filmed' in either walk<br>G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7<br>G9 mask definition differs: window_subtracted differs between the paired keyframes |
| left rear fender | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| right front fender | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| right rear fender | G1, G3, G4 | G1 not filmed: the panel is not 'filmed' in either walk<br>G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe<br>G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7 |
| hood | G4, G8 | G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7<br>G8 hood IoU: hood pair with whitened mask IoU under 0.8 |
| roof | G1, G6 | G1 not filmed: the panel is not 'filmed' in either walk<br>G6 side uncertain: side_uncertain, or front_sign null, on either walk |
| front bumper | G4 | G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7 |
| left front wheel | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| left rear wheel | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| right front wheel | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| headlights | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| tail lights | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| mirrors | G1, G3, G4, G6 | G1 not filmed: the panel is not 'filmed' in either walk<br>G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe<br>G4 mask IoU: whitened mask IoU of the paired keyframes under 0.7<br>G6 side uncertain: side_uncertain, or front_sign null, on either walk |
| windshield | G1 | G1 not filmed: the panel is not 'filmed' in either walk |
| rear window | G1, G3 | G1 not filmed: the panel is not 'filmed' in either walk<br>G3 blurry keyframe: sharpness below 0.4 of the candidates' max on either keyframe |

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## Method and limits

Every verdict on the panel pages was decided by code from the numbers below. The thresholds were frozen into this run under config hash `89dbc3e3f49f0557e848db8e744a679f5a8615224e031eacf26d813416e9407f`. Rebuilding the derived files from the stored model answers and these thresholds gives the same bytes when the line below says so.

| | |
|---|---|
| **rebuild** | event chain verified to the head on page 1 and any original still in quarantine re-hashed at sheet time; run walkaround verify to compare a rebuild |
| **storage** | Model requests were sent with storage off and the models accepted it; nothing was kept after the run. Files API upload deleted: 2026-11-01T05:24:23.620Z; quarantine deleted: 2026-11-01T05:24:23.620Z. |
| **Hertz thresholds** | Example thresholds Hertz published (REV 7.2025), illustrative only, quoted from its Damage Charges support page saved 2026-10-26: a body scratch is billable when it cannot be covered by a credit card (2 in x 3.5 in), or when scratches over 1 in together exceed that area; a dent with a diameter over 1 in, or multiple dents; a wheel, rim or hubcap scratch or scuff over 1 in; any crack or chip in glass. Paint transfer or scuffs that do not break the paint are listed as Ordinary Wear and not charged. |

### Eval numbers

No eval run is attached to this sheet; the README carries the eval tables with their N.

### Calibration

| | |
|---|---|
| **tau_pre (already there below)** | `0.4` |
| **tau_eye (needs your eye below)** | `0.8` |
| **tau_fix (Stays Put fixed below)** | `0.08` |
| **sigma_frame / sigma_spark / sigma_view / sigma_within** | `None / 0.117 / 0.051742 / 0.127931` |
| **calibrated** | no: tau_pre hit its cap, so the matcher is not calibrated for this run |

### Model calls (from runs/ledger.jsonl)

| kind | calls | dollars | quantity |
|---|---|---|---|
| sam_image | 184 | 0.4600 | 184 images |
| sam_video | 64 | 2.4896 | 12448 frames |
| spark | 71 | 0.3506 | 138260 tokens |
| spark_open | 1 | 0.0043 | 2670 tokens |
| voice | 1 | 0.0009 | 19.76 s |
| **total** | 321 | 3.3054 | |

### What was blurred

- pickup: SAM image checks on 27 keyframes found a plate on 15 and faces on 0; the display proxy was pixelated from the plate and face tracks
- return: SAM image checks on 42 keyframes found a plate on 12 and faces on 0; the display proxy was pixelated from the plate and face tracks

### What was deleted and when

| kind | target | sha256 | bytes | deleted at | reason | receipt |
|---|---|---|---|---|---|---|
| file | media/pickup.proxy.mp4 | a5327b83fd9f | 16354164 | 2026-11-01T05:21:42Z | unblurred proxy deleted at blur.done | |
| file | media/return.proxy.mp4 | 095b907b280c | 15180664 | 2026-11-01T05:21:44Z | unblurred proxy deleted at blur.done | |
| files_api | file-4747963342191598 | a5327b83fd9f | | 2026-11-01T05:24:23.620Z | run end (pickup proxy) | {'deleted': True, 'id': 'file-4747963342191598', 'object': 'file'} |
| files_api | file-1010094108804044 | 095b907b280c | | 2026-11-01T05:24:23.620Z | run end (return proxy) | {'deleted': True, 'id': 'file-1010094108804044', 'object': 'file'} |
| quarantine | quarantine/pickup.orig | 25f68a8992c9 | 96182720 | 2026-11-01T05:24:23.620Z | run end | |
| quarantine | quarantine/return.orig | f49b432f2932 | 107096877 | 2026-11-01T05:24:23.620Z | run end | |

### What the seal does not prove

- which car was filmed, unless the plate was kept on this sheet at the renter's request
- that the walk was not staged before the upload; the seal dates the bytes, not the events in them
- anything about the pickup condition when the seal was made after the return

The seal is offered as a process record under FRE 901(b)(9): the same inputs through the same code give the same output, and the method page states that code's inputs.

### Thresholds frozen into this run

| | |
|---|---|
| claim.align_landmark_fit | False |
| claim.dark_v_max | 0.15 |
| claim.drawn_box_inside_saturated_max | 0.1 |
| claim.drawn_box_perimeter_saturated_min | 0.3 |
| claim.drawn_box_saturation_min | 0.5 |
| claim.full_panel_area_min | 0.03 |
| claim.full_panel_border_max | 0.15 |
| claim.landmark_fit_min_landmarks | 3 |
| claim.landmark_fit_residual_max_frac | 0.1 |
| claim.low_res_px | 480 |
| claim.on_panel_frac | 0.025 |
| claim.passenger_side_is | right |
| claim.photo_flag_caps_one_walk_only | True |
| claim.r_claim_full_panel | 0.1 |
| claim.r_claim_two_landmarks | 0.2 |
| claim.roi_pad_frac | 0.25 |

| | |
|---|---|
| claim.roi_region_max_panel_frac | 0.4 |
| claim.sam_vote_requests | 14 |
| claim.scanner_style_saturation_max | 0.08 |
| claim.view_flag_caps | True |
| cost.d1_budget_usd | 25 |
| cost.pair_estimate_usd_10fps | 4.9 |
| cost.pair_estimate_usd_15fps | 6.2 |
| cost.pickup_check_estimate_usd | 2.6 |
| cost.sam_image_usd | 0.0025 |
| cost.sam_video_min_frames | 50 |
| cost.sam_video_usd_per_frame | 0.0002 |
| cost.voice_usd_per_hour | 0.18 |
| coverage.area_only_box_area_frac_min | 0.01 |
| coverage.c_present_floor | 0.5 |
| coverage.filmed_area_frac_min | 0.03 |
| coverage.filmed_completeness_min | 0.85 |
| coverage.filmed_sharpness_rel_min | 0.4 |
| coverage.glimpsed_c_percentile | 25 |
| coverage.odometer_ask_once | True |
| coverage.topup_order | ['wheels', 'lower_bumpers', 'rest'] |
| dedup.cluster_radius_units | 0.2 |
| dedup.collinear_angle_deg | 10 |
| dedup.collinear_gap_units | 0.1 |
| dedup.count_mode_new_point_min_units | 0.1 |
| dedup.droplet_max_size_units | 0.03 |
| dedup.droplet_min_marks | 8 |
| dedup.group_min_marks | 3 |
| dedup.group_radius_units | 0.5 |
| dedup.road_spray_band_bottom_frac | 0.25 |
| dedup.road_spray_min_marks | 3 |
| dedup.small_mark_px | 12 |
| dedup.small_mark_units | 0.03 |
| eval.billable_buckets_mm | [10, 20, 25.4] |
| eval.false_mark_radius_units | 0.35 |
| eval.recall_radius_units | 0.25 |
| eval.report_pairs | ['p2xr2', 'cross', 'multi_day'] |
| eval.threshold_fit_pairs | ['pseudo', 'p1xr1'] |
| geometry.bumper_arc_fallback_tau_factor | 1.5 |
| geometry.carried_moments_check_frames | 5 |
| geometry.carried_moments_residual_max_units | 0.02 |
| geometry.hood_depth_thirds | 3 |
| geometry.perspective_residual_center_max | 0.08 |
| geometry.perspective_residual_corner_max | 0.25 |
| geometry.unit_cm_along_door | 35 |
| geometry.unit_cm_vertical_door | 29 |
| geometry.whiten_affine_invariance_tol | 1e-06 |
| matching.band_radius_factor | 0.5 |
| matching.chance_match_density_formula | pi * tau_pre^2 / 12 * mark_density |

| | |
|---|---|
| matching.cm_per_unit_for_print | 35 |
| matching.compatible_types | ['scratch', 'scuff', 'chip'] |
| matching.hood_mask_iou_min | 0.8 |
| matching.margin_min_tau_pre_factor | 0.5 |
| matching.mask_iou_min | 0.7 |
| matching.next_keyframe_pair | True |
| matching.pseudo_pair_percentile | 90 |
| matching.pseudo_pair_share_min | 0.9 |
| matching.ratio_test_min | 1.5 |
| matching.sharpness_rel_min | 0.4 |
| matching.sigma_spark | 0.117 |
| matching.size_cost_factor | 0.1 |
| matching.tau_eye_factor | 2.0 |
| matching.tau_pre0_floor | 0.25 |
| matching.tau_pre0_sigma_factor | 3.535534 |
| matching.tau_pre_cap | 0.4 |
| matching.tight_pair_second_look | True |
| matching.type_cost_compatible | 0.1 |
| matching.type_cost_incompatible | 0.4 |
| matching.wheel_min_marks_for_shift | 2 |
| matching.wheel_radius_tolerance | 0.1 |
| matching.wheel_size_factor_max | 2.0 |
| media.anchor_clip_frames | 180 |
| media.anchor_clip_half_window_s | 6.0 |
| media.chunk_bytes | 8388608 |
| media.chunk_route_max_body_bytes | 9437184 |
| media.chunks_in_flight | 3 |
| media.clip_crf | 18 |
| media.clip_frames | 91 |
| media.clip_half_window_s | 1.5 |
| media.clip_prompt_frame | 45 |
| media.clip_size_px | 1008 |
| media.files_expires_after_s | 3600 |
| media.inline_data_url_max_bytes | 52428800 |
| media.other_route_max_body_bytes | 1048576 |
| media.proxy_codec | libx264 |
| media.proxy_crf | 20 |
| media.proxy_fps | 15 |
| media.proxy_fps_fallbacks | [10, 30] |
| media.proxy_max_edge | 960 |
| media.proxy_preset | medium |
| media.source_fps_assumed | 30 |
| media.upload_max_bytes | 1073741824 |
| media.upload_warn_bytes | 524288000 |
| media.wav_rate | 24000 |
| nouns.d1_text_prompts | ['scuff'] |
| nouns.door_handle_image_requests_per_pair | 24 |
| nouns.image_only | ['door handle'] |

| | |
|---|---|
| nouns.lanes | ['car', 'license plate', 'face', 'car door', 'bumper', 'wheel', 'hood', 'trunk lid', 'fender', 'roof', 'headlight', 'tail light', 'side mirror', 'windshield', 'rear window', 'car window'] |
| nouns.never_prompt | ['front bumper', 'rear bumper'] |
| nouns.privacy | ['license plate', 'face'] |
| nouns.topup_always | ['car', 'license plate', 'face'] |
| nouns.voice_keywords | ['car', 'car door', 'bumper', 'wheel', 'hood', 'trunk lid', 'fender', 'roof', 'headlight', 'tail light', 'side mirror', 'windshield', 'rear window', 'car window', 'scuff', 'scratch', 'chip', 'dent', 'crack', 'tape', 'odometer'] |
| orchestrator.baseten_rpm | 30 |
| orchestrator.baseten_semaphore | 1 |
| orchestrator.caps_if_no_card_ceiling.pair_runs_per_day | 0 |
| orchestrator.caps_if_no_card_ceiling.pickup_runs_per_day | 3 |
| orchestrator.daily_card_ceiling_cents | 2000 |
| orchestrator.decode_workers | 2 |
| orchestrator.health_probe_failures_to_pause | 3 |
| orchestrator.health_probe_interval_s | 300 |
| orchestrator.judging_card_ceiling_cents | 5000 |
| orchestrator.max_concurrent_runs | 2 |
| orchestrator.media_workers | 2 |
| orchestrator.pair_runs_per_day | 1 |
| orchestrator.pdf_timeout_s | 60 |
| orchestrator.pickup_runs_per_day | 6 |
| orchestrator.priority_wave_size | 6 |
| orchestrator.queue_eta_from_last_runs | 3 |
| orchestrator.rate_limit_backoff_base_s | 0.5 |
| orchestrator.rate_limit_retries | 6 |
| orchestrator.replay_fixed_total_s_estimate | 35 |
| orchestrator.replay_recorded_delay_cap_ms | 2000 |
| orchestrator.request_retries | 1 |
| orchestrator.request_timeout_s | 240 |
| orchestrator.run_deadline_s | 720 |
| orchestrator.run_spend_ceiling_cents | 800 |
| orchestrator.sam_image_semaphore | 16 |
| orchestrator.sam_video_semaphore_global | 32 |
| orchestrator.sam_video_semaphore_per_run | 16 |
| orchestrator.spark_semaphore | 12 |
| orchestrator.sse_ping_s | 15 |
| orchestrator.sweeper_interval_s | 3600 |
| orchestrator.volume_min_free_bytes | 524288000 |
| panels.anchor_absorb_iou_min | 0.5 |
| panels.anchor_max_frames_per_panel | 3 |
| panels.anchor_negative_point_seam_offset_box_widths | 0.1 |
| panels.anchor_pass_coverage_below | 0.7 |
| panels.area_floor_default | 0.02 |
| panels.area_floor_small | 0.005 |
| panels.border_frac_keyframe_max | 0.15 |

| | |
|---|---|
| panels.c_floor | None |
| panels.car_box_containment_min | 0.8 |
| panels.car_mask_containment_min | 0.8 |
| panels.crop_long_edge_px | 1280 |
| panels.crop_pad_frac | 0.1 |
| panels.direction_min_px_per_frame | 2.0 |
| panels.direction_phase_corr | False |
| panels.fast_mask_decode | True |
| panels.fragment_gap_split_frames | 2.5 |
| panels.fragment_gap_split_s | 0.5 |
| panels.fragment_min_frames | 2 |
| panels.fragment_min_s | 0.5 |
| panels.glare_s_max | 40 |
| panels.glare_v_min | 250 |
| panels.keyframe_box_pass_candidates | 9 |
| panels.keyframe_decode_budget_s_per_video | 60 |
| panels.keyframe_min_gap_s | 0.5 |
| panels.keyframe_panels_in_flight | 4 |
| panels.keyframe_score_exponents.area | 1.0 |
| panels.keyframe_score_exponents.completeness | 1.0 |
| panels.keyframe_score_exponents.frontality | 1.0 |
| panels.keyframe_score_exponents.one_minus_glare | 1.0 |
| panels.keyframe_score_exponents.sharpness | 1.0 |
| panels.keyframe_top_k | 3 |
| panels.keyframe_top_k_under_budget | 2 |
| panels.merge_area_ratio | [0.6, 1.6] |
| panels.merge_aspect_tolerance | 0.2 |
| panels.merge_gap_max_s | 1.0 |
| panels.merge_pred_dist_box_widths | 0.5 |
| panels.prompt_box_pad_px | 2 |
| panels.side_vote_margin_min | 1 |
| panels.side_vote_weights.R1 | 2 |
| panels.side_vote_weights.R2 | 1 |
| panels.side_vote_weights.R3 | 1 |
| panels.side_vote_weights.R4 | 1 |
| panels.small_nouns | ['side mirror', 'headlight', 'tail light', 'license plate', 'face'] |
| panels.twin_check.adjacent_lid_sizes | 0.25 |
| panels.twin_check.margin_min | 3 |
| panels.twin_check.min_frames | 5 |
| panels.twin_check.mirror_far_lid_sizes | 1.5 |
| panels.twin_check.mirror_near_lid_sizes | 1.0 |
| panels.twin_check.neighbour_majority | 2.0 |
| panels.twin_check.neighbour_share_min | 0.5 |
| panels.twin_check.photo_margin_min | 2 |
| panels.twin_check.plate_inside_min | 0.8 |
| panels.twin_check.plate_share_min | 0.5 |
| panels.twin_check.presence_margin | 0.4 |
| panels.twin_check.weights.glass | 1 |

| | |
|---|---|
| panels.twin_check.weights.lights | 1 |
| panels.twin_check.weights.mirror | 2 |
| panels.twin_check.weights.plate | 2 |
| privacy.box_blocks_across | 4 |
| privacy.files_delete_retries | 3 |
| privacy.image_check_nouns | ['license plate', 'face'] |
| privacy.image_checks_in_flight | 16 |
| privacy.keep_plate_default | False |
| privacy.pixel_block_px_at_4k | 20 |
| privacy.run_retention_hours | 24 |
| privacy.track_box_dilation_frac | 0.1 |
| privacy.track_box_dilation_px | 24 |
| privacy.track_box_union_s | 0.1 |
| privacy.track_carry_fit_s | 0.4 |
| privacy.track_carry_grow_per_s | 0.5 |
| privacy.track_carry_s | 3.0 |
| privacy.track_carry_speed_max_frac_per_s | 1.0 |
| privacy.track_flow_cover_max_frac | 0.1 |
| privacy.track_flow_cover_path | True |
| privacy.track_flow_grow_per_s | 0.2 |
| privacy.track_gap_bridge_s | 0.5 |
| privacy.unblurred_temp_max_minutes | 30 |
| ruler.axis_tilt_correction_max_deg | 30 |
| ruler.body_scuff_ordinary_wear | True |
| ruler.bound_px_per_side | 2 |
| ruler.camera_hfov_deg | 70 |
| ruler.d1_plate_wheel_ppi_agreement | 0.15 |
| ruler.mark_min_px | 8 |
| ruler.plate_c_min | 0.5 |
| ruler.plate_hw_min | 0.35 |
| ruler.plate_long_side_in | 12 |
| ruler.plate_min_px | 60 |
| ruler.plate_tilt_max_deg | 30 |
| ruler.ruler_max_distance_m | 0.6 |
| ruler.threshold_body_scratch_in | 3.5 |
| ruler.threshold_dent_in | 1.0 |
| ruler.threshold_wheel_in | 1.0 |
| ruler.tire_diameter_formula | D = 2 * (SW * AR / 2540 + RD / 2) |
| ruler.view_flag_sizes | True |
| ruler.wheel_c_min | 0.5 |
| ruler.wheel_min_px | 120 |
| sheet.footage_credit | True |
| spark.confidence_levels | ['low', 'medium', 'high'] |
| spark.d1_jaccard_pass | 0.7 |
| spark.d1_sigma_pass_frac_of_crop_width | 0.02 |
| spark.effort | low |
| spark.grid | 1000 |
| spark.grid_slack | 20 |

| | |
|---|---|
| spark.image_long_edge_px | 1280 |
| spark.input_tokens_per_1280_crop_estimate | 1400 |
| spark.jpeg_quality | 90 |
| spark.landmarks.bumper | ['plate_corner_tl', 'plate_corner_tr', 'plate_corner_bl', 'plate_corner_br', 'fog_lamp', 'grille_corner', 'sensor'] |
| spark.landmarks.door | ['handle', 'lock', 'mirror_base', 'window_front_corner', 'window_rear_corner', 'wheel_arch', 'lower_trim', 'badge'] |
| spark.landmarks.hood | ['badge', 'washer_nozzle', 'front_corner_left', 'front_corner_right'] |
| spark.landmarks.wheel | ['hub', 'rim_lip', 'valve'] |
| spark.landmarks_max | 4 |
| spark.mark_types | ['scratch', 'scuff', 'chip', 'dent', 'crack', 'dirt', 'reflection', 'unsure'] |
| spark.max_output_tokens | 4000 |
| spark.model | muse-spark-1.3 |
| spark.retry_max_output_tokens | 8000 |
| spark.schema_names | ['marks_v1', 'count_v1', 'look_v1', 'seed_v1', 'claim_name_v1', 'timeline_v1', 'note_prefill_v1'] |
| spark.schema_retry_count | 1 |
| spark.second_look_engine | spark_open |
| spark.second_looks_per_pair_max | 25 |
| spark.size_classes | ['S', 'M', 'L'] |
| spark.store | False |
| spark.stream_when_expected_over_s | 30 |
| staysput.b1_floor | 0.15 |
| staysput.b2_floor | 0.05 |
| staysput.cap_enabled | False |
| staysput.check_box_covered_min | 0.9 |
| staysput.clip_blur_dilation_frac | 0.3 |
| staysput.clip_frames_max | 181 |
| staysput.control_blob_max_frac | 0.15 |
| staysput.control_blob_min_px | 9 |
| staysput.curvature_boundary_units | 0.15 |
| staysput.d1_pass_box_share | 0.8 |
| staysput.d1_pass_drift_max | 0.08 |
| staysput.d1_reflection_drift_min | 0.16 |
| staysput.highlight_s_percentile | 10 |
| staysput.highlight_v_percentile | 95 |
| staysput.max_clips_per_walk | 16 |
| staysput.max_objects_per_clip | 16 |
| staysput.min_bbox_px_at_4k | 12 |
| staysput.min_drift_frames | 10 |
| staysput.neighbour_dedupe_iou | 0.5 |
| staysput.neighbour_fixed_below | 1.5 |
| staysput.neighbour_handle_half_h_frac | 0.02 |
| staysput.neighbour_handle_half_w_frac | 0.08 |
| staysput.neighbour_mark_min_clip_px | 16 |
| staysput.neighbour_near_half_window_s | 0.5 |

| | |
|---|---|
| staysput.neighbour_part_max_window_frac | 0.5 |
| staysput.neighbour_residual_floor_px | 0.5 |
| staysput.neighbour_sliding_above | 2.5 |
| staysput.neighbour_track_share_min | 0.8 |
| staysput.neighbour_window_margin | 1.15 |
| staysput.neighbour_window_max_growth | 2.0 |
| staysput.neighbour_window_max_panel_frac | 1.5 |
| staysput.neighbours_enabled | True |
| staysput.neighbours_max | 12 |
| staysput.neighbours_min | 4 |
| staysput.neighbours_target | 6 |
| staysput.no_track_box_share_below | 0.8 |
| staysput.panned_inconclusive_share_caption_threshold | 0.5 |
| staysput.reflection_control_from_hood_glare | True |
| staysput.sign_agreement_min | 0.8 |
| staysput.sign_pair_gap_frames | 5 |
| staysput.sign_pairs_min | 8 |
| staysput.sign_panel_dx_min_px_per_frame | 2.0 |
| staysput.sliding_factor | 2.0 |
| staysput.tau_fix_floor | 0.08 |
| staysput.tau_fix_sigma_factor | 3.0 |
| staysput.window_frac_of_panel_width | 0.3 |
| staysput.window_mark_margin | 1.5 |
| staysput.window_min_px | 400 |
| staysput.window_share_overlap_min | 0.5 |
| voice.chip_min_ms | 100 |
| voice.max_audio_s | 600 |
| voice.max_body_bytes | 33554432 |
| voice.mode | ENDPOINTING |
| voice.model | muse-voice-transcribe-1.0 |
| voice.modes_allowed | ['PUSH_TO_TALK', 'ENDPOINTING', 'DIARIZATION'] |
| voice.pin_fallback | turn_start_panel |
| voice.seeds_per_walk_max | 16 |

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

## Appendix A certification

```
Renter's certification

I, [name], state that:
1. I recorded the videos listed below myself, at pickup on [date] and at return on [date], of the vehicle
under
   rental agreement [number].
2. I uploaded them to Walkaround without editing them, and the sha256 hashes on page 1 match the files on
my phone.
3. I understand this sheet is evidence support, not legal advice, and that it was produced by code from
my videos.

- pickup: uco3d_7336_narrated.mp4 (sha256 25f68a8992c9...)
- return: uco3d_50152-5241-46685.mp4 (sha256 f49b432f2932...)

Signed: _____   Date: _____

This is a signed statement, not a notarized affidavit.
```

## Appendix B glossary

```
Glossary

already there: the same mark was found at pickup within tau_pre of its return position (rule L3).
needs your eye: a human should look; the pair sat in the band between tau_pre and tau_eye, or a cap (K1
to K15) applied (L4).
new or unseen: a return mark with no pickup mark near it; new, or not filmed at pickup (L2).
not seen at return: a pickup mark with no return mark near it (L1).
not comparable: a panel gate failed (G1, G3 to G10); both best frames still show with the reason (L6).
view flagged: Muse Spark said a frame shows water, dirt or glare. That is a model's judgment, so it no
longer blocks a panel (gate G2 did that until Oct 31). A mark seen in one walk only is capped at needs
your eye (K15); a mark in the same place in both walks keeps its label, with a note.
tight pair: a pair inside tau_pre is already there because the mark sits in the same place in both walks,
whatever Muse Spark called it. So Spark's type (dirt, reflection or unsure, K4) and its low confidence
(K12) are notes on that pair, and the second look compares the two crops and may lower it (K7). If the
second look gives no answer, K4 and K12 cap the pair at needs your eye as before. A mark seen in one walk
only, or a pair in the band, keeps K4 and K12 as caps.
keyframe pair: each panel is compared on one pickup keyframe and one return keyframe. When that pair
fails a gate of the pair (G3, G4, G5, G8 or G9), the panel's other keyframe pairs are tried in a fixed
order (clean views first, then the two keyframe scores, then keyframe number) and the first that passes
every gate is used. The panel page says which pair was used and what each pair tried failed. G1, G6, G7
and G10 belong to the panel, so no other pair can pass them.
units: panel-normalized distance; 1 unit is about 35 cm along a door (the about-cm beside it uses that).
tau_pre, tau_eye: the two distance thresholds, calibrated from the run's own repeat frames; printed on
the method page.
Stays Put: a check that a mark moves with the panel through a short clip; a reflection slides, a scratch
stays put.
sha256: the fingerprint of a file; one changed byte changes it.
sealed: the manifest of hashes was written and offered for time stamping before any model saw the video.
config hash: the fingerprint of every threshold the code used; it changes when a threshold changes.
rule id: the row of the decision table that produced a label; G gates a panel, K caps a mark, L names a
label, E is a claim exit.
```

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

## Appendix C verify it yourself

```
Verify it yourself (5 steps; none needs Walkaround's server)

1. Hash the video file on your phone or computer and compare it with page 1:
   shasum -a 256 <your video file>     (Linux: sha256sum <your video file>)
   pickup: 25f68a8992c980976df1352c0d2894eca301a13b576bcac397e61952e1440efb
   return: f49b432f2932d4a9ffeb59881c992c5054e09a2d7d6b816a41d5dd1df9c8646b
2. Read the OpenTimestamps proof (pip install opentimestamps-client):
   ots info bundle/manifest.json.ots
3. Verify it against the Bitcoin block, or drop the .ots file on https://opentimestamps.org :
   ots verify bundle/manifest.json.ots
4. Verify the RFC 3161 token from freetsa (2026-11-01T05:19:42Z) with freetsa's certificates (download
https://freetsa.org/files/cacert.pem and https://freetsa.org/files/tsa.crt):
   openssl ts -verify -data bundle/manifest.json -in bundle/manifest.json.tsr -CAfile cacert.pem -
untrusted tsa.crt
5. Re-hash every file in the bundle and rebuild the derived files from the stored model answers:
   walkaround verify pairC_core3
```

---

Walkaround run pairC_core3. Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0, https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS.

## Letter: write to the company first

```
Write to the company first

[date]

To: [rental company], damage claims
Re: rental agreement [number], picked up [date], returned [date]

I received a damage charge of [amount] for [description]. Before anything else I ask for the full damage
report,
the written repair estimate and every photograph or scan the company holds for this rental, including the
pickup
inspection images. Hertz's own support page says customers can request more images.

I recorded a walkaround video at pickup and at return. The files were hashed and sealed before any
analysis:
- pickup: uco3d_7336_narrated.mp4, 96182720 bytes, sha256
25f68a8992c980976df1352c0d2894eca301a13b576bcac397e61952e1440efb, camera time not present, received 2026-
11-01T05:19:38.420Z
- return: uco3d_50152-5241-46685.mp4, 107096877 bytes, sha256
f49b432f2932d4a9ffeb59881c992c5054e09a2d7d6b816a41d5dd1df9c8646b, camera time not present, received 2026-
11-01T05:19:38.533Z
- sealed at 2026-11-01T05:19:38.673Z
- Walkaround run pairC_core3; the panel-by-panel evidence sheet is attached
- the frames on the sheet: Car videos: uCO3D (Liu et al., Meta, 2025), CC BY 4.0,
https://github.com/facebookresearch/uco3d . License: https://creativecommons.org/licenses/by/4.0/ . We
cropped, re-encoded and blurred plates in these clips. Narration: ours, Piper TTS. License:
https://creativecommons.org/licenses/by/4.0/

The attached sheet shows the panel in question at pickup and at return with the marks drawn on both
frames. The
sheet is evidence support, not legal advice, and I ask that it be read next to your own images.

If the charge stands after you have reviewed it, I will obtain an alternative written estimate within 10
business
days. Card network rules (Visa Core Rules 13.3 and 13.5; 15 USC 1666i) require that I first try to
resolve the
charge with you; this letter is that attempt. Please reply within 14 days to [email].

[name]
```

---